

INDEX

SL No.	Name of the Experiment	Page No.	Date of Experiment	Date of Submiss
1	Even or odd	1		
2	Leap year or not	2		
3	Factorial of a Number	3		
4	Swap two Integers	4		
5	Add two numbers	5		
6	Celcius to Fahrenheit			
7	Prime number or not	7		
8	Print prime no within a range	8		
9	child process	10		

1. Write a shell program to check the given number is even or odd:

Input:

```
echo "Enter the number: "  
read n  
r=$((n % 2))  
if [ $r -eq 0 ]  
then  
echo "$n is Even."  
else  
echo "$n is Odd."  
fi
```

output

Enter the number:

7

7 is odd.

Teacher's Signature : _____

2. Write a shell program to check the given year is leap year or not:

Input:

```
echo "Enter the year: "  
read y  
b='expr $y % 4'  
if [ $b -eq 0 ]  
then  
echo "$y is a leap year."  
else  
echo "$y is not a leap year."
```

output

Enter the year:

2000

2000 is a leap year

Teacher's Signature : _____

3. Write a shell program to find the factorial of a number.

Input:

```
echo "Enter a number."
```

```
read n
```

```
i = `expr $n - 1`
```

```
p = 1
```

```
while [ $i -ge 1 ]
```

```
do
```

```
n = `expr $n \* $i`
```

```
i = `expr $i - 1`
```

```
done
```

```
echo "The factorial of the given number is:"
```

output

Enter a number

5

The factorial of the given number is: 120.

Teacher's Signature : _____

4. Write a shell program to swap two integers.

Input:

```
echo "Enter two numbers."
```

```
read a b
```

```
temp = $a
```

```
a = $b
```

```
b = $temp
```

```
echo "After swapping"
```

```
echo $a $b
```

output:

Enter two numbers .

5 7

After swapping

7 5

Teacher's Signature : _____

5. Write a shell program to add two numbers

Input:

```
read -p "Enter first number: " num1
read -p "Enter second number: " num2
sum = $(($num1 + num2))
echo "Sum is: $sum"
```

Output:

```
Enter first number: 85
Enter second number: 19
Sum is 104
```

Teacher's Signature : _____

7. Check a number if it's prime or not.

Input:

```
echo -e "Enter number:"
```

```
read n
```

```
for ((i=2; i<= $n/2; i++))
```

```
do
```

```
ans=$(( n%i ))
```

```
if [ $ans -eq 0 ]
```

```
then
```

```
echo "$n is not a prime number."
```

```
exit 0
```

```
fi
```

```
done
```

```
echo "$n is a prime number."
```

output

Enter number:

47

47 is a prime number."

Teacher's Signature : _____

8. Print all the prime numbers within a range:

Input:

```

prime = 0
echo "Enter the range"
read n
echo "Prime number between 1 to $n is: "
echo "1"
echo "2"
for ((i=3; i<=n; i++))
do
    for ((j=i-1; j>=2; j--))
    do
        if [ `expr $i % $j` -ne 0 ] ; then
            prime = 1
        else
            prime = 0
            break
        fi
        i = `expr $j - 1`
    done
    if [ $prime -eq 1 ] ; then
        echo $i
    fi
    i = `expr $i + 1`
done

```

Teacher's Signature : _____

output

Enter the range

50

Prime number between 1 to 50 is

2

3

5

7

11

13

17

19

23

29

31

37

41

43

47

Teacher's Signature : _____

Input:

```
#include <stdio.h>
#include <unistd.h>
int main() {
    int pr_id;
    pr_id = fork();
    if (pr_id == 0) {
        printf("\n from child with pid=%d, ppid=%d
        and fork return value %d\n", getpid(),
        getppid(), pr_id);
    }
```

```
    else if (pr_id > 0) {
        printf("\n from parent with pid=%d, ppid=%d
        and fork return value %d\n", getpid(),
        getppid(), pr_id);
    }
```

```
    else {
        printf("\n error in fork pid=%d, ppid=%d
        and fork return value %d\n", getpid(),
        getppid(), pr_id);
    }
```

Output

From parent with pid=3380, ppid=3283. and fork_ return value = 3381 ← From child with pid=3381, ppid=3380 and fork_return value = 0.

Teacher's Signature :